RIFT | Five Apertus 1.5 Readiness Findings

Hack Apertus 2026 - Track 1A - Consolidated technical report - 14 October 2026

Abstract. Five documented model failures span temporal document interpretation, Swiss cantonal voting rules, Swiss QR-bill constraints, FIX trading-event classification and synthetic release authorization decisions. All were evaluated on the Public AI-hosted `swiss-ai/apertus-v1.5-8b` identifier at temperature zero. Observed errors are narrow, task-specific, and not claims of actual-world operational compromise.

1 | Method and evaluation boundaries

We use frozen scenario prompts, a separately specified deterministic correct-answer oracle, archived provider completions, and contrastive controls. Counts below refer to hosted response records or evaluated classifications, not population prevalence. An API model identifier is not independent attestation of the precise hosted checkpoint. A fresh, credentialed inference is distinct from replay of archived provider evidence.

2 | Findings at a glance

ID / class	Observed hosted behavior	Ground truth / scope
F01 Temporal attribution	9 archived records; 8 unique completion IDs; 4 unique failing IDs.	Ignore amendment not yet effective for queried entity/date.
F02 Swiss civic eligibility	2 wrong exact German completions; paraphrase and federal controls passed.	Eligible Glarus cantonal voter at age 17.
F03 QR-bill pairing	Expected A,C -> A,B,D; fresh expected B,D -> A,D; isolated control passed.	Ordinary IBAN+SCOR and QR-IBAN+QRR valid.
F04 FIX 4.4 execution	Trade events returned 150=8 and 150=D instead of 150=F; ablation corrected event type.	New trade ExecType=F; filled OrdStatus=2.
F05 Release attestation	16 labels / 3 completions; 5 false ALLOW, 2 false BLOCK.	Two distinct eligible non-revoked signers required.

3 | F01 and F02 - temporal and jurisdictional knowledge

F01. An effective-date extraction task incorrectly selected future-active business amendments as present evidence. Researcher archive contains nine recorded requests, eight unique provider completion IDs and four unique failed IDs. A later-as-of-date and fresh entity constrain the interpretation; the error is not a demonstrated production harm. **F02.** A 17-year-old otherwise eligible Swiss citizen domiciled in Glarus was incorrectly labelled ineligible to vote in a cantonal referendum. Two distinct completions of the same German prompt failed; paraphrased and federal-election controls passed. Cantonal and federal eligibility cannot be conflated.

4 | F03 and F04 - domain constraints

F03. QR-bill validation confused legal IBAN/reference-type pairings on two synthetic multirow fixtures, despite independently valid identifier formats; one isolated QRR control passed. Neither funds transfer nor account authorization occurred. **F04.** FIX 4.4 executions that completed an order were labelled Rejected (150=8) or Restated (150=D), not Trade (150=F); the ablation removing a distracting message-code annotation corrected ExecType but the model still used the textual status "Filled" in place of wire value 39=2. This is a protocol reasoning and format error, not demonstrated tag copying.

5 | F05 - synthetic release-attestation decisions

The policy requires at least two **distinct** eligible signers for the same release, digest, production scope and gate time, with signer-specific revocation handling. Across three hosted outputs and sixteen decisions the model issued five false ALLOW and two false BLOCK recommendations. Controlled tests show duplicate grants counted as independent approvals and unrelated revocations affecting valid releases; a fresh shuffled ledger showed further false ALLOWs. No live deployment was made.

6 | Reproduction contract and limitations

The private GitHub submission repository requires five schema-valid issue JSON files, original archived request/response envelopes and deterministic per-finding oracles. Judges execute `make run` from its root using Docker. Archived-evidence audit is labelled as archival verification; any new hosted-model inference is optional, credentialed and non-default.

Prior art includes known factual, temporal and constraint-reasoning error classes. Results do not establish system-wide prevalence, deliberate discrimination, executed financial operations, release bypass, or independent checkpoint SHA. Severity, breadth, novelty and validity grades belong to judges. Detailed prompts, controlled comparisons, provider metadata and disclosure boundaries are in each per-finding technical dossier.

References

Hack Apertus Project Template, Track 1A README and findings.schema (GitHub, 2026).

Swiss Confederation, Voting rights; Constitution of the Canton of Glarus, Art. 56.

SIX Group, Swiss QR-bill Implementation Guidelines, v2.3 (2025).

FIX Trading Community, FIX 4.4 ExecType and ExecutionReport specification.

RIFT archived provider evidence and individual F01-F05 dossiers (13-14 October 2026).